UNIVERSITI TEKNOLOGI MALAYSIA

Project Task 4

Programme: Bachelor of Computer Science (Data Engineering)

Subject: SECR1213 Network Communications

Session-Sem: 2024/2025-1

Section: 02

Group name: Router ranger

Group Member	Matric Number
Choh Jing Yi	A23CS0296
Neo Li Xin	A23CS0253
Pravinraj a/l Sivabathi	A23CS0171
Ahmad Ziyaad bin Mohd Abbas	A23CS0206

Index

4.1 Identify the work areas on floor plan

Figure 4.1.1: Work Area at Ground Floor

Work Area 1 shown in figure 4.1.1 is a student meeting room. It is eight meters wide and nine meters long. It is perfect for cooperative group discussions and gatherings because it has an oval table with six chairs around it. A workstation and a projector are among the useful equipment in the room that supports student activities and helps to improve group project discussions.

Work Area 2 shown in figure 4.1.1 is student lounge measures 10 meters in width and 14 meters in length. Students can use this lounge to relax or study before class. It is equipped with eight workstations for academic tasks, along with a rest area featuring three comfortable sofas for relaxation. Along with a printer to let students print study materials, the area also has a selection of books for study purposes. Furthermore, a Wi-Fi access point is set up to offer dependable and simple network connectivity, ensuring that students stay connected and productive while using the student lounge.

Work Area 3 and 4 as shown in Figure 4.1.1 are both general labs spanning 14 meters in length and 10 meters in width. These general labs are intended for student use during classes. Each lab is equipped with a total of 60 PC workstations providing students hands-on learning and collaborative projects.

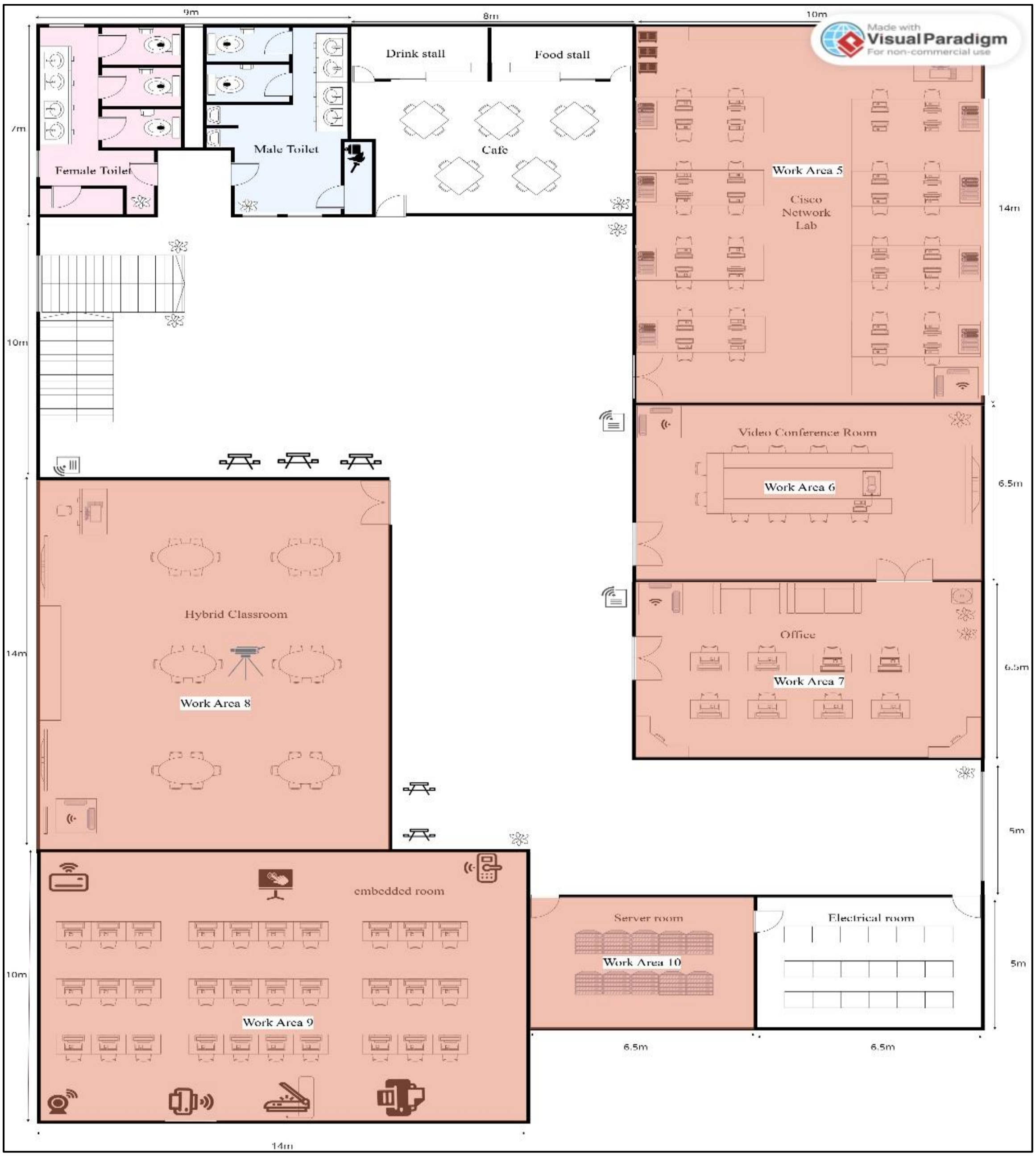


Figure 4.1.2: Work Area at First Floor

The Cisco Network Lab for work area 5 shown in figure 4.1.2 is 10 meters wide and 14 meters long. It is in the upper-right corner of the first level. There are four rows of tables with four computer workstations each on the left and right sides of the lab, providing students lots of space for learning. There is also an instructor's desk at the top-right corner, equipped with a workstation. Additionally, a Wi-Fi access point is installed in the bottom-right corner for wireless connectivity.

Video Conference Room is located at work area 6 shown in figure 4.1.2 which designed for meetings and virtual collaboration. The room measures 6.5 meters in length 10m in width and is organized around a large central conference table with 8 chairs. At end of the table, there is a designated workstation equipped with a computer and projector. The room also includes a large display screen on the wall collaborate with projector to facilitate presentations and virtual meetings. Additionally, a Wi-Fi access point is installed on the left side of the room ensure high speed Wi-Fi connectivity for video conferencing.

For the work area 7 shown in figure 4.1.2, it is a lecturer's office with dimensions of 6.5 meters in length and 10 meters in width. The office contains eight workstations, arranged in two rows of four, each equipped with a chair and computer setup. There are two sofas positioned at the top of the room, providing a comfortable seating area. Additionally, a Wi-Fi access point is installed in the top-left corner to ensure wireless connectivity.

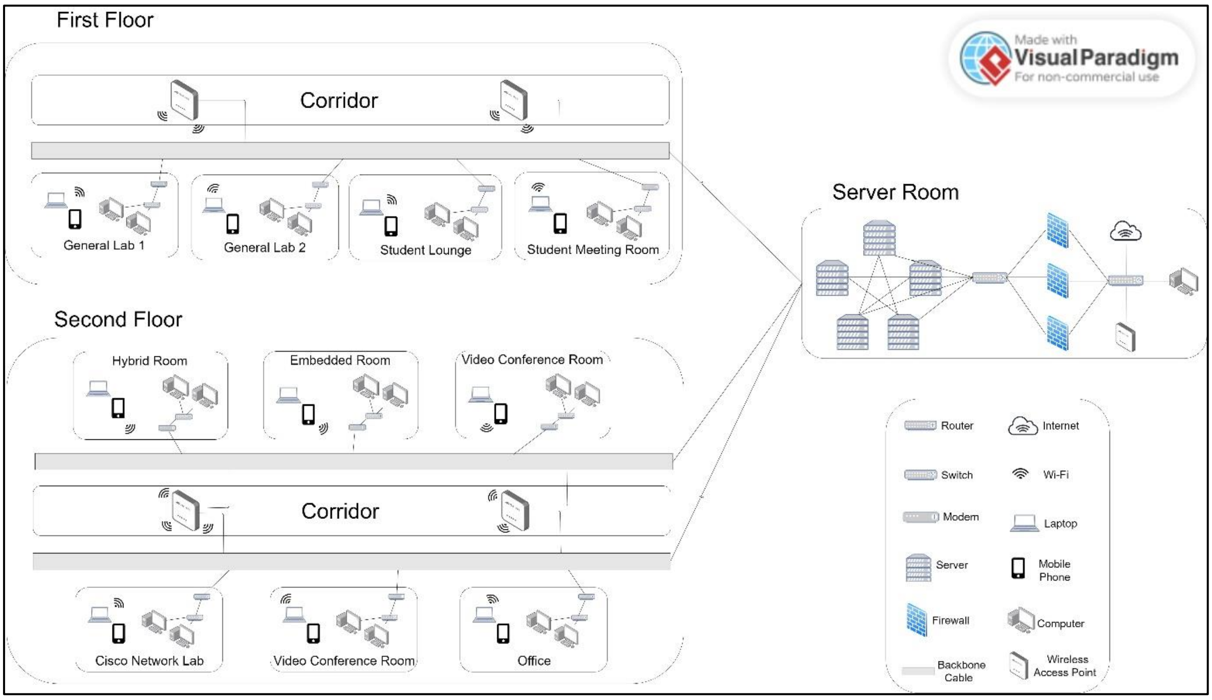
Work area 8 shown in figure 4.1.2 is hybrid classroom. The hybrid classroom must be setup for best performance and efficiency for teaching and learning. A video conferencing camera set up in the middle of the room allows for distant participation and hybrid instruction. In addition, the classroom has a teacher's workstation, a presentation display screen, and a Wi-Fi access point for constant online connectivity.

Work area 9 shown in figure 4.1.2 is embedded lab designed to facilitate learning in embedded systems and IoT. It has 30 workstations, ensuring that students have the resources they need to complete practical tasks. The lab contains a variety of IoT equipment, such as digital sensors (temperature, motion, and pressure sensors), Wi-Fi, and Bluetooth communication devices, which allow for real-time data collection and wireless networking. IoT is supported by additional peripherals like as webcams, printers, and scanners.

Work area 10 shown in figure 4.1.3 is server room. The room measures 6.5 meters in width and 5 meters in length, and it is equipped with arranged racks that have servers, network switches, routers, and other necessary hardware. The server room is designed to allow high-capacity data processing, storage and connection making it the foundation of IT operations.

4.2 The distribution and connection of devices

4.2.1 Network Distribution Diagram



4.2.1.1 Network Distribution Diagram

Figure 4.2.1.1 shows the entirety of the network distribution within the faculty of computing. This network distribution diagram aims to visually depict the layout and connectivity of the devices in each work area, including routers, switches, wireless access points and end devices. This diagram will also show the cabling routes for horizontal and vertical connections, making it clear what the logical and physical topology of the network is.

Figure 4.2.1.2 and Figure 4.2.1.3 shows the network distribution of the work areas of the first floor, including two general labs, student lounge and student meeting room. The workstations are connected to the router and then to the modem through ethernet cable. This is to ensure seamless connectivity and better network performance as compared to wireless connection. Lastly the modem is connected through optic fibre to the wireless access point. The end devices such as mobile devices and laptop are connected wirelessly to the wireless access point which provides the Internet IP addresses.

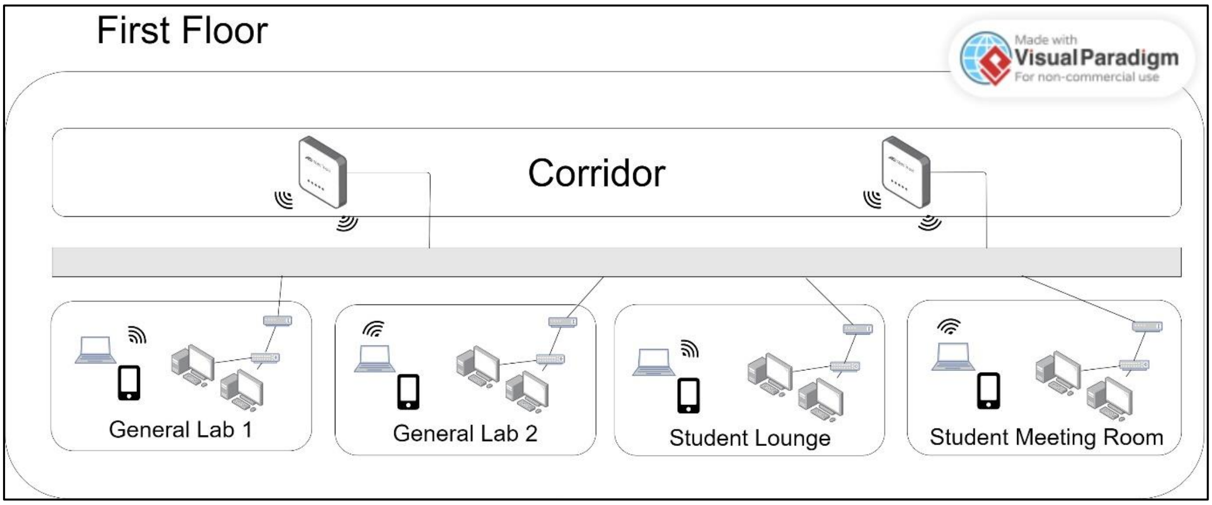


Figure 4.2.1.2 Network Distribution Diagram – First Floor

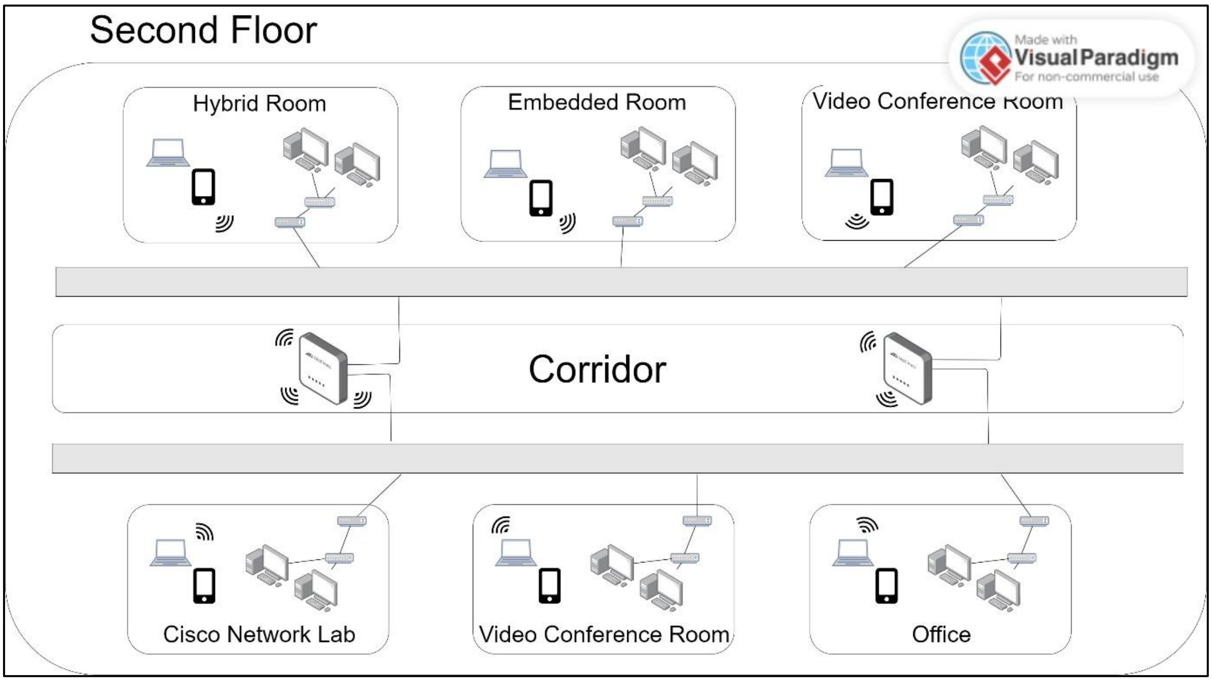


Figure 4.2.1.3 Network Distribution Diagram – Second Floor

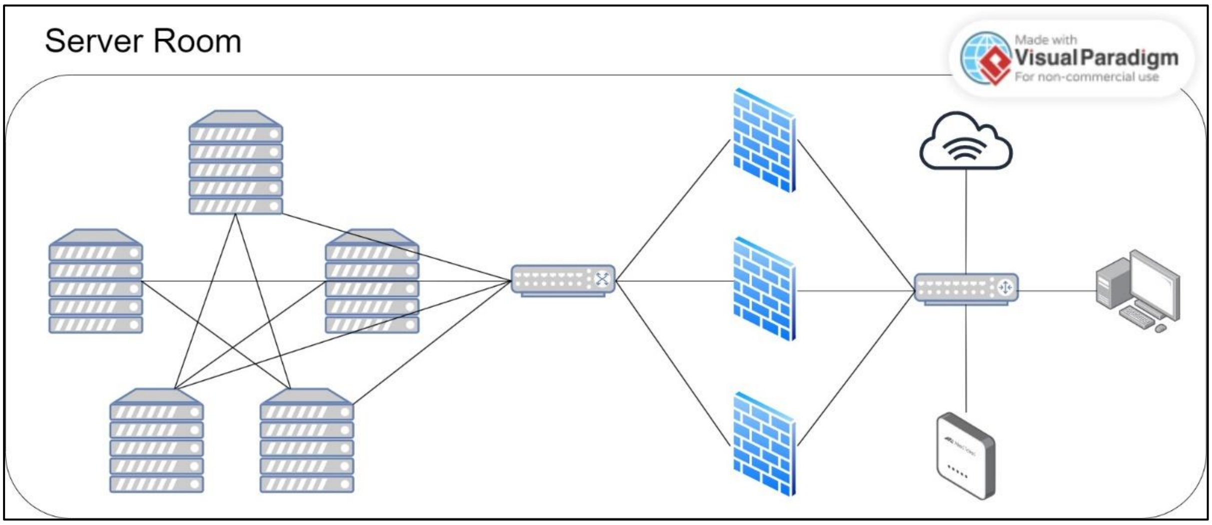


Figure 4.2.1.4 Network Distribution Diagram – Server Room

As seen in Figure 4.2.1.4, the servers are interconnected using a mesh topology. This network distribution can increase reliability and longevity of the server's services as the other server is still able to connect to the Internet even when one server is malfunction or under reparation. This design also promotes scalability as new server can be added without interrupting the existing network, making it better for future growth.

The faculty of computing's servers are mainly used as a centralized storage for research data, student work, course materials and faculty resources. Besides, it acts as a host core network infrastructure to manage traffic between the internal network and the Internet.

In order to maintain network security, the servers are directed to the firewalls through switches before entering the internal network infrastructures through the router. Cisco networking lab, embedded lab, hybrid lab and general labs are highly dependent on the servers as they deliver quality networking with low latency to ensure seamless connection data system management.

4.2.2 Room Level Network Distribution Diagram

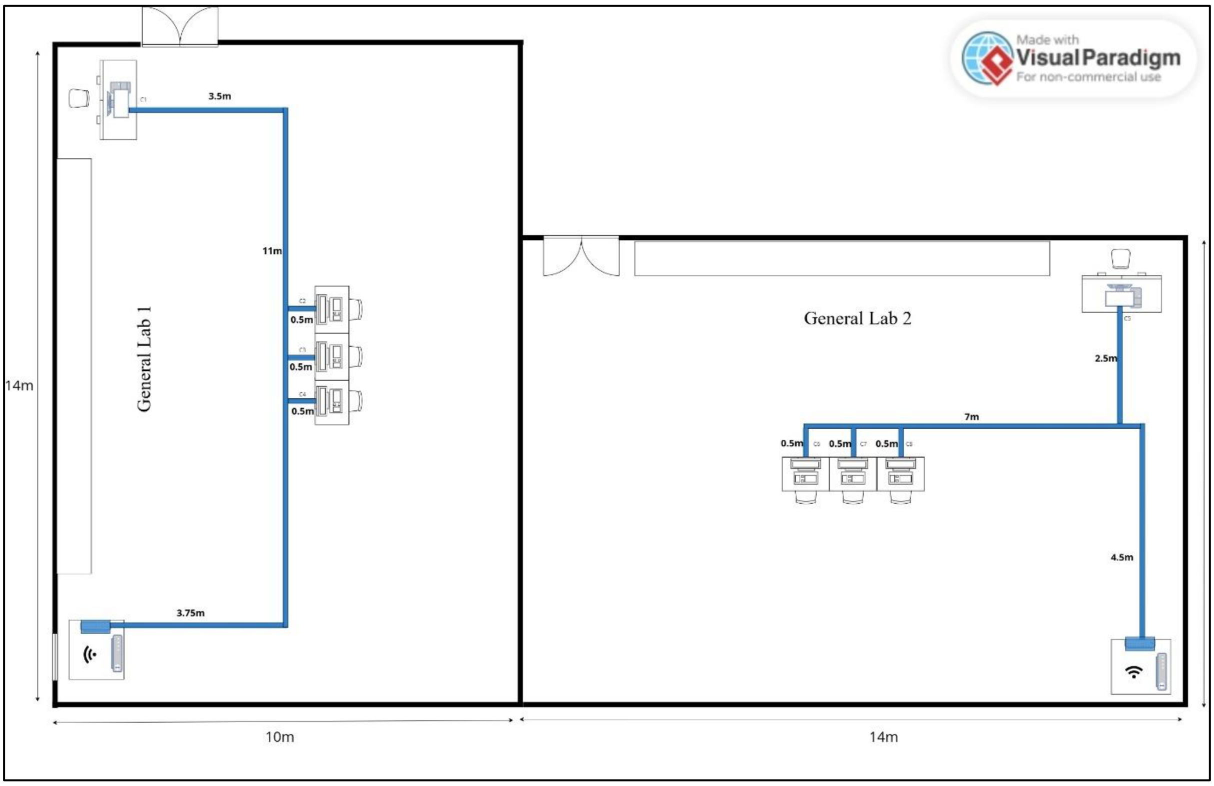


Figure 4.2.2.1: General Lab

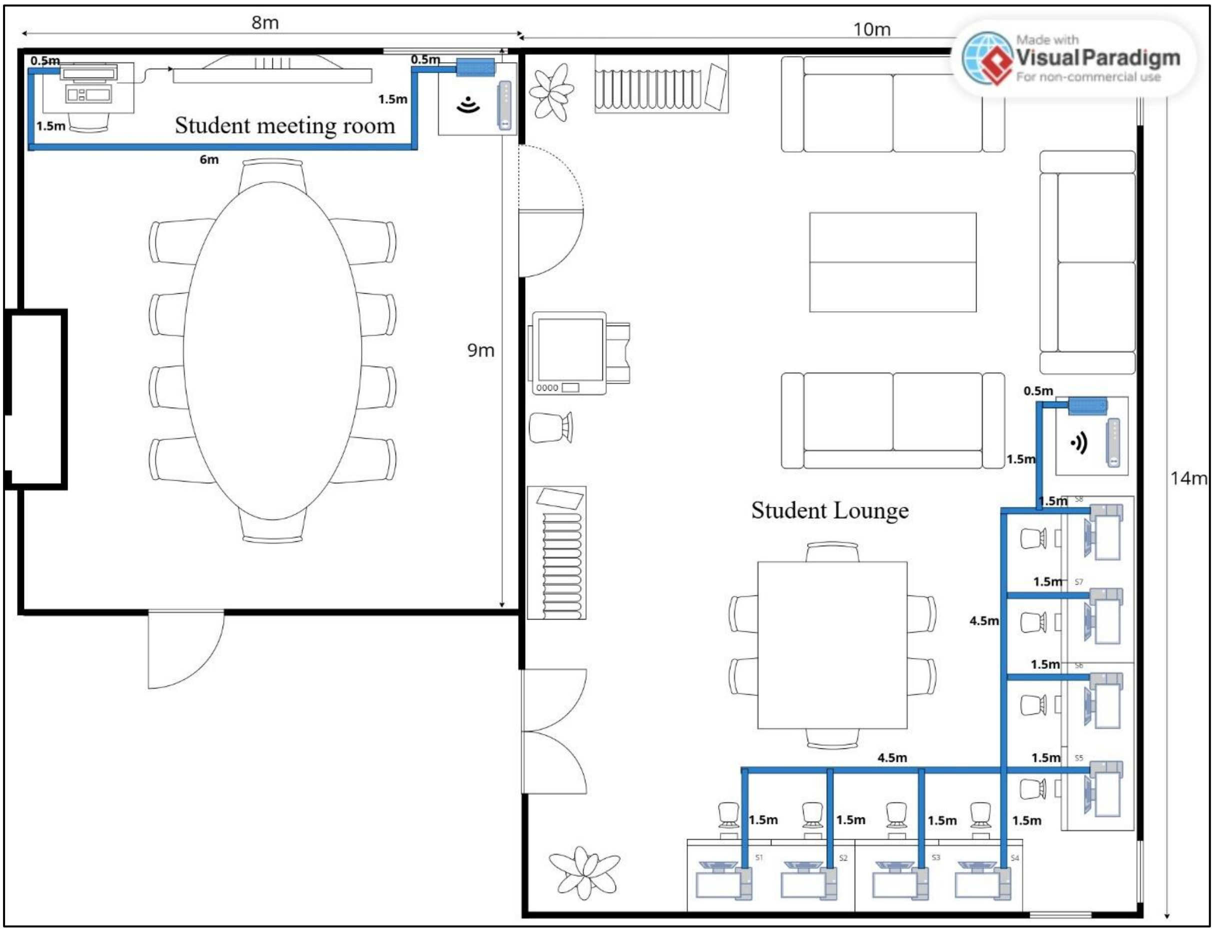


Figure 4.2.2.2: Student Meeting Room and Student Lounge

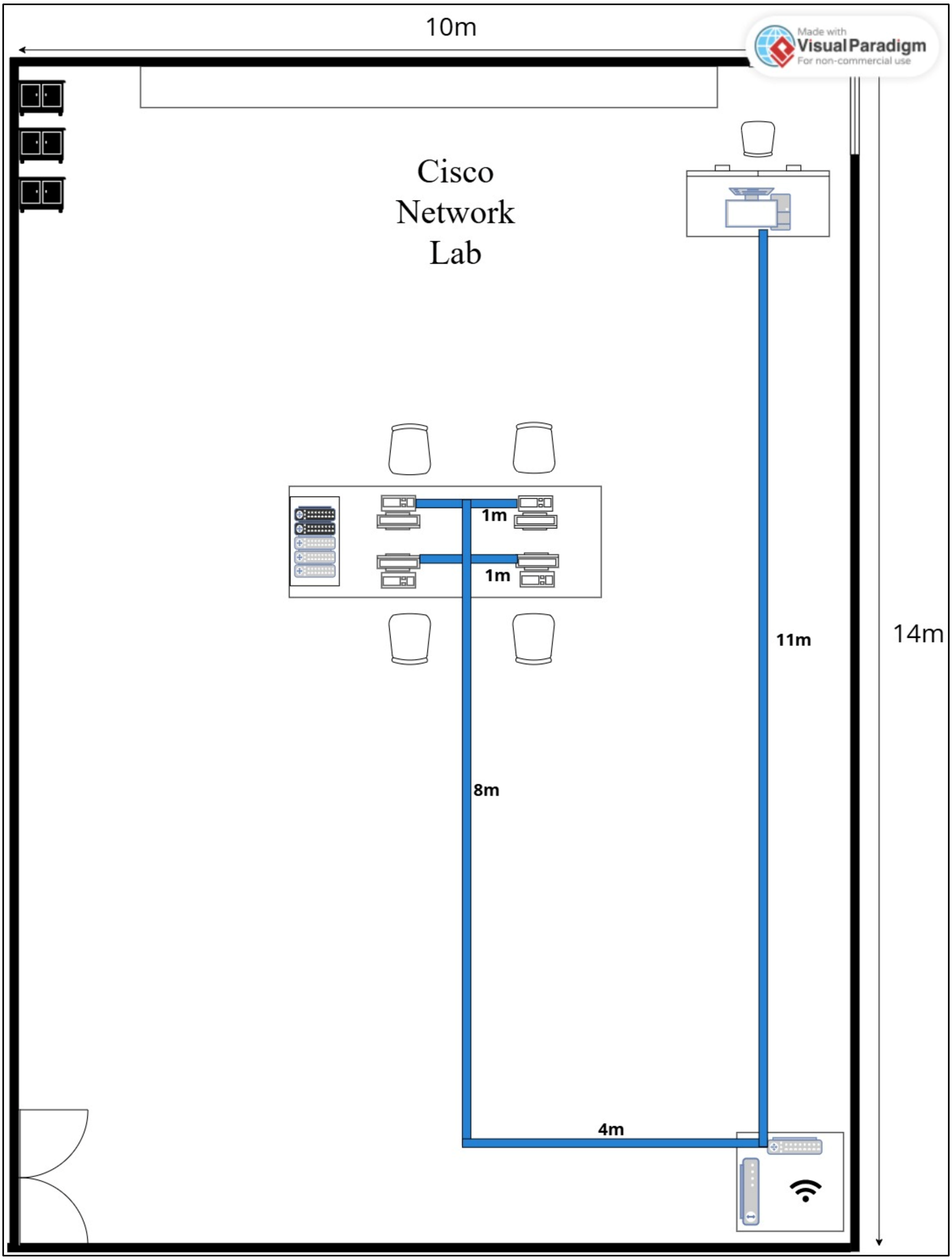


Figure 4.2.2.3: Cisco Network Lab

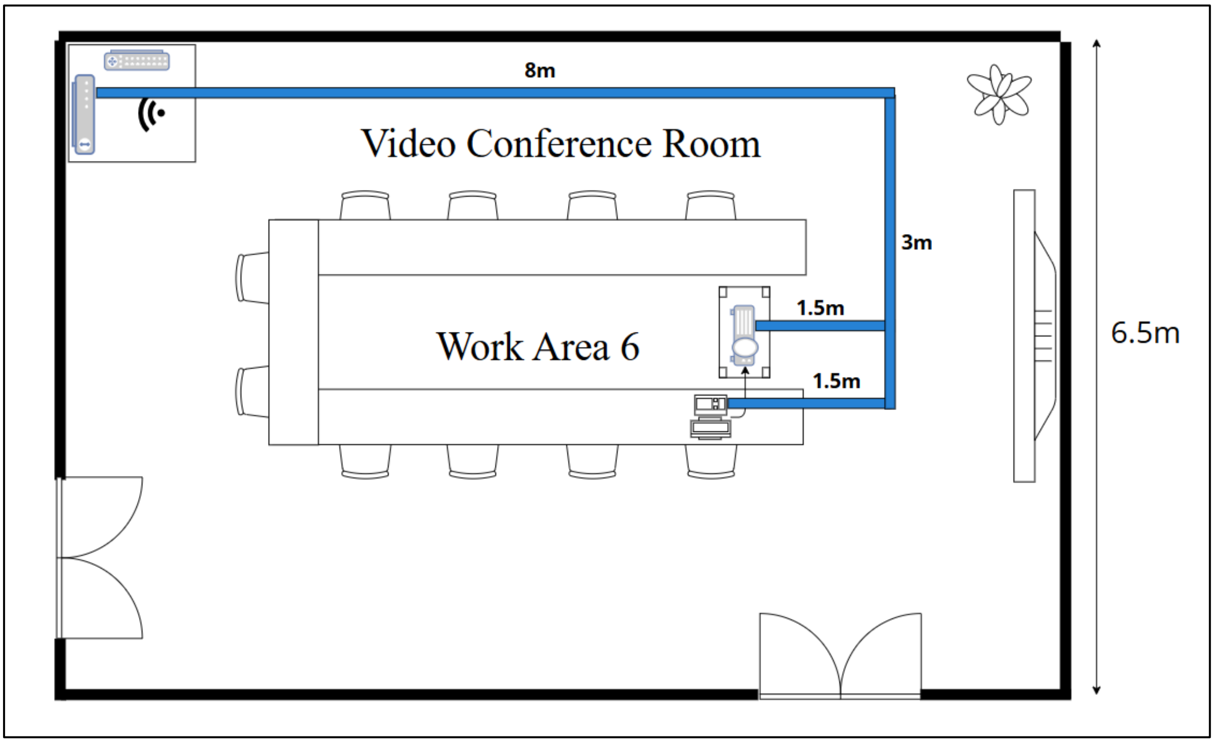


Figure 4.2.2.4: Video Conference Room

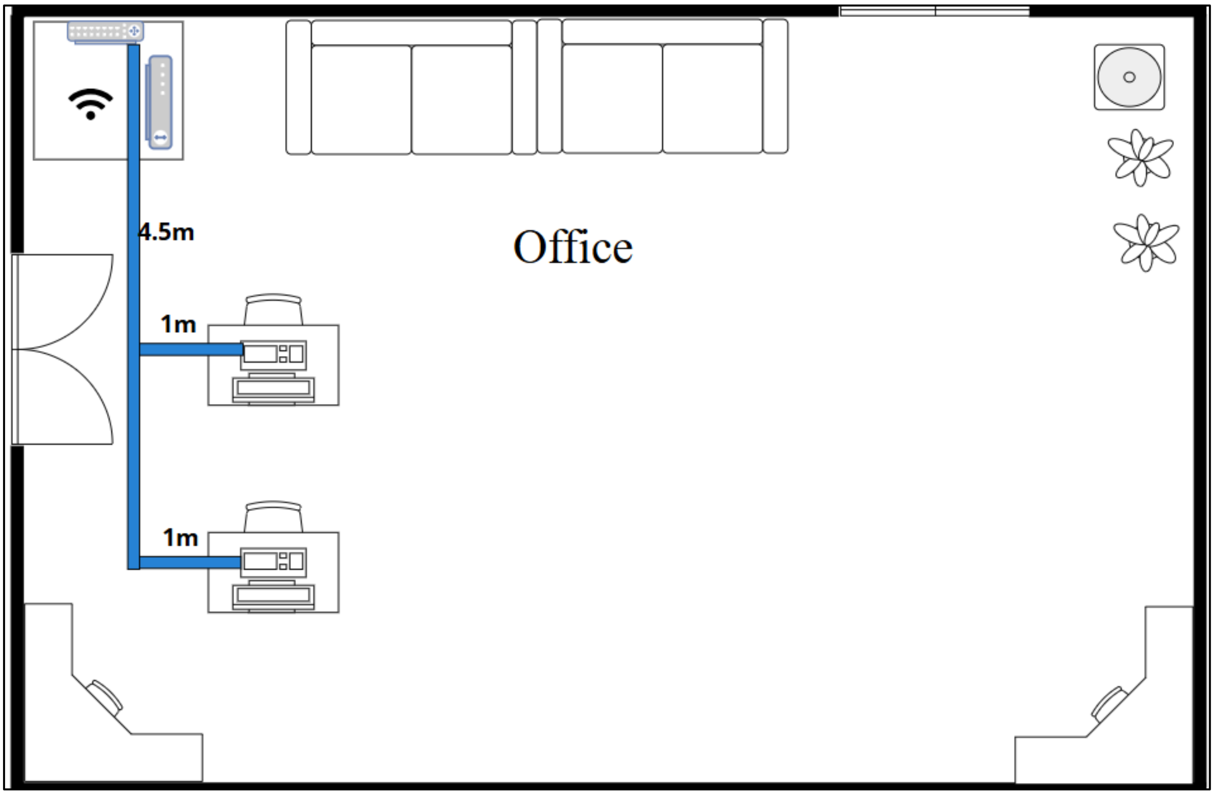


Figure 4.2.2.5: Office

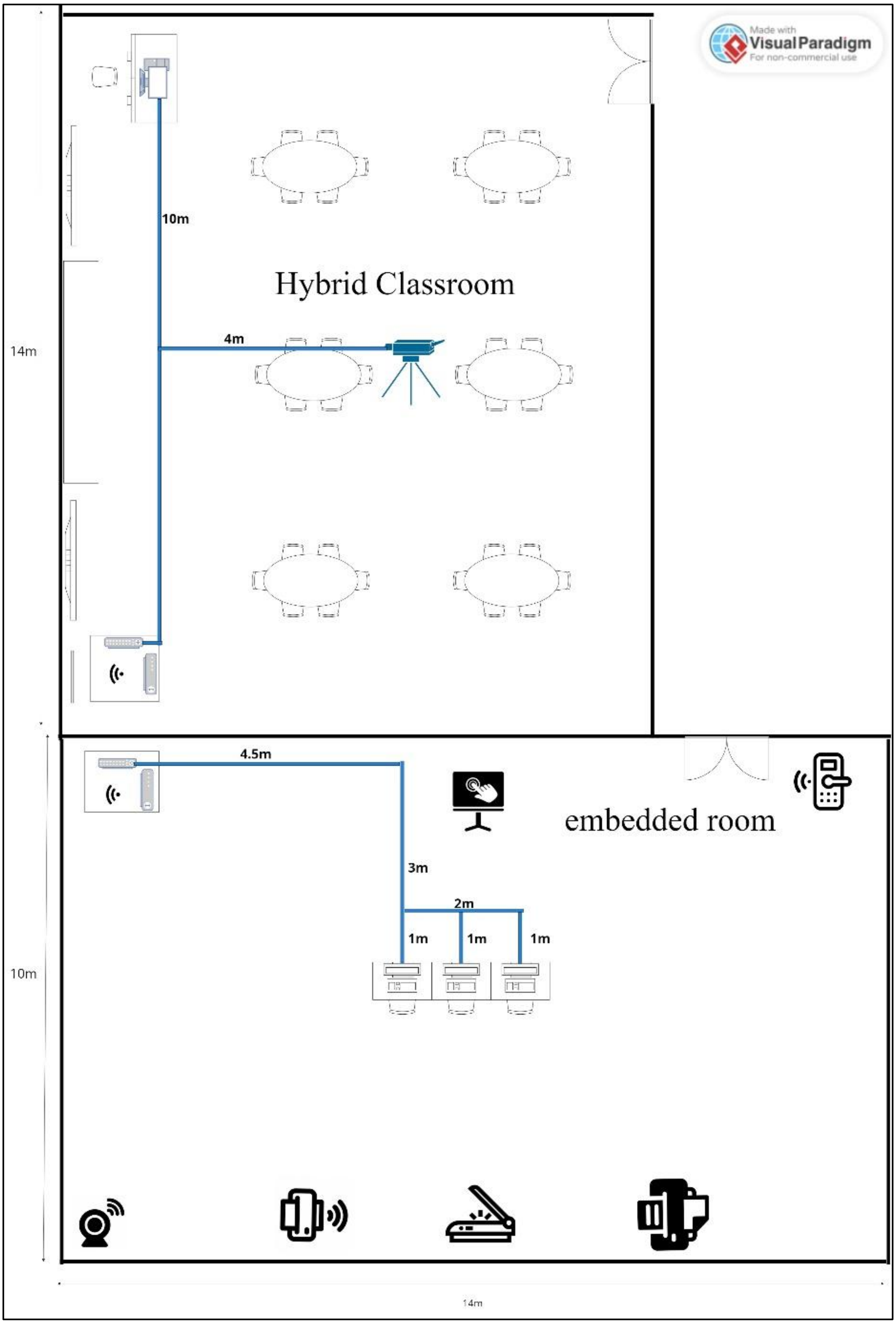


Figure 4.2.2.6: Hybrid Classroom and Embedded Room

4.2.3 Floor Diagram

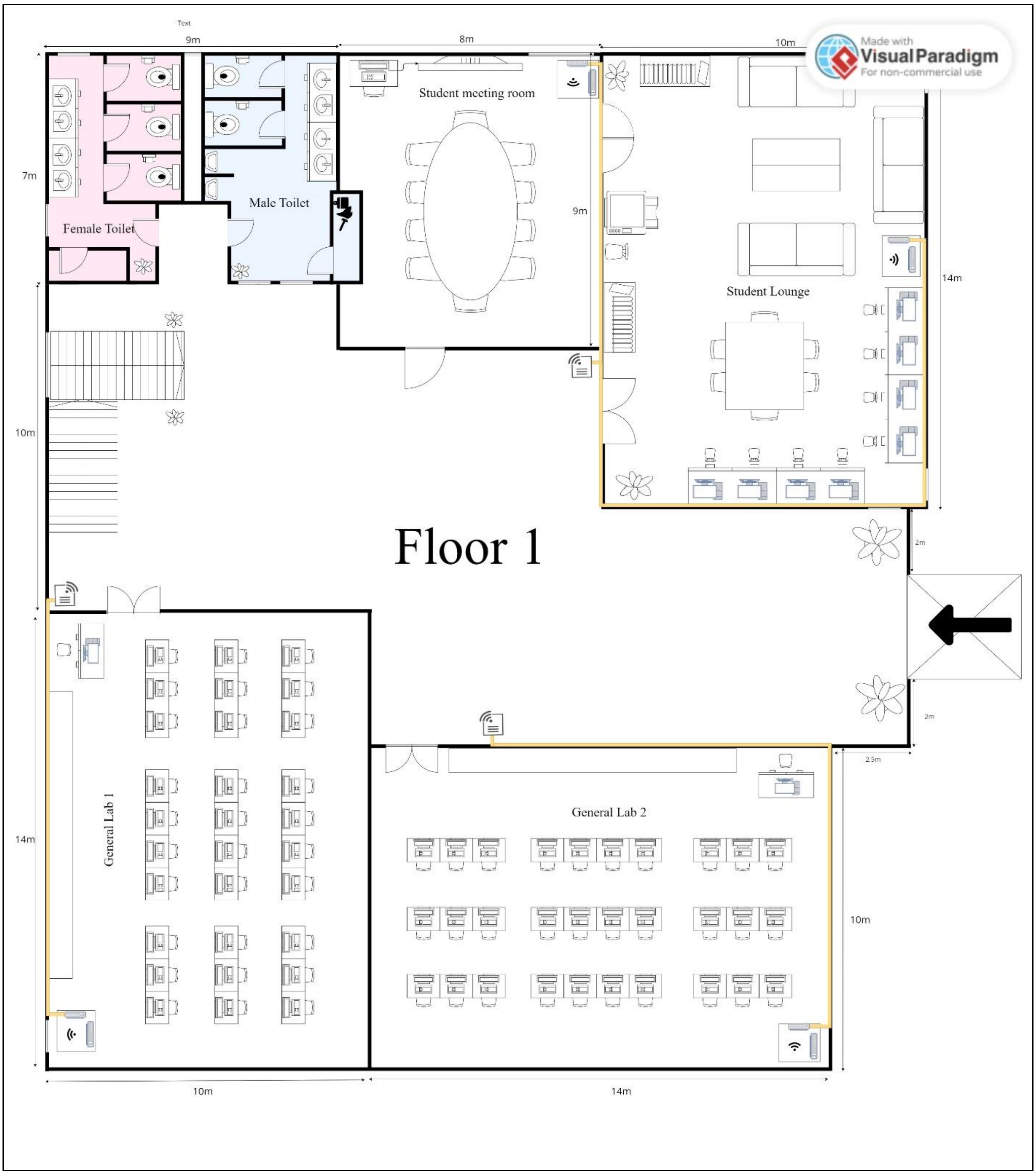


Figure 4.2.3.1 Floor Diagram – First Floor

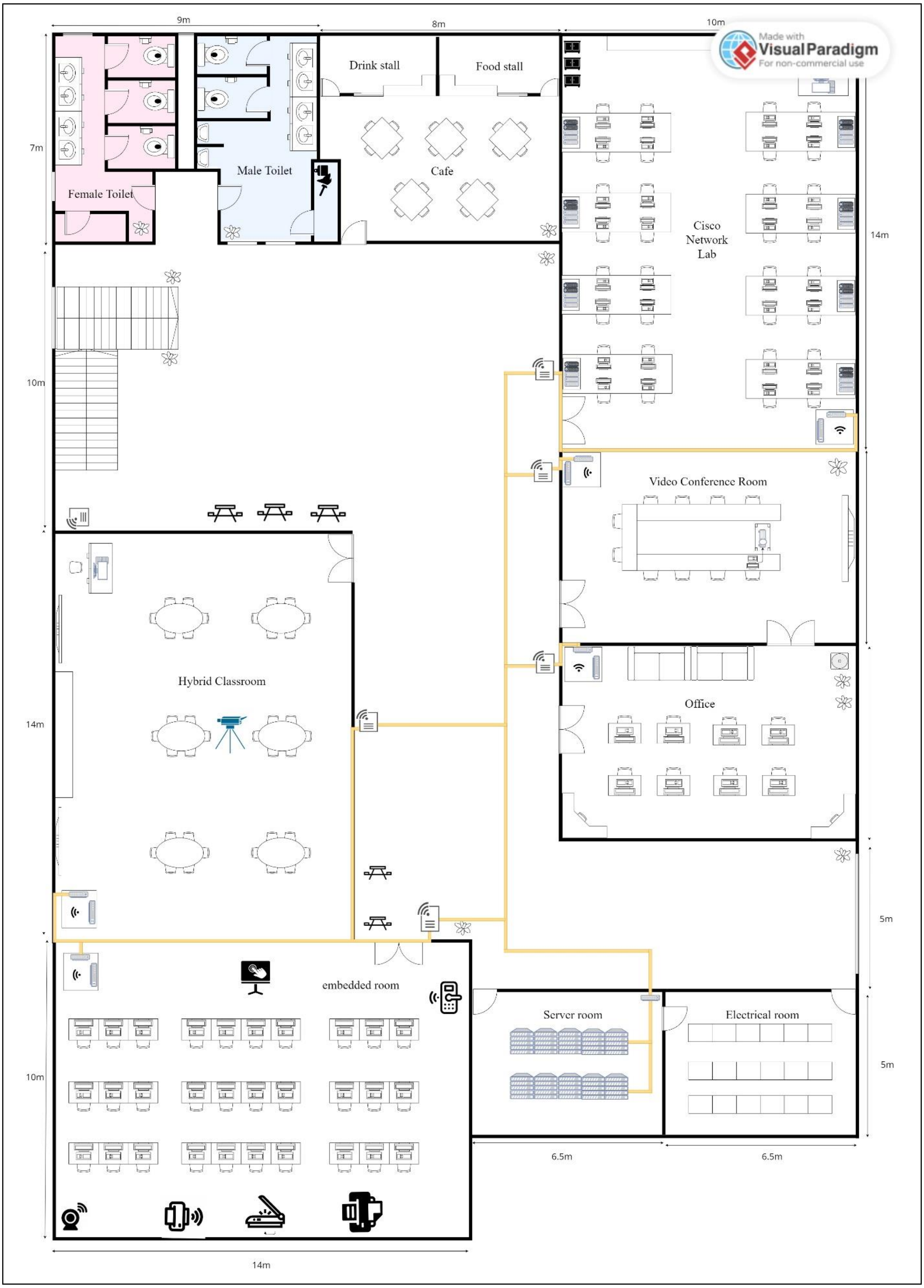


Figure 4.2.3.2 Floor Diagram – Second Floor

4.3 Number of connections, patch cords and switch ports needed

Work Area	Number of Devices	Type of Devices	Number of Ports Required
Student Meeting Room	6 Workstations	PCs, Wi-Fi Access Point	7
Student Lounge	8 Workstations, 1 Printer	PCs, Printer, Wi-Fi	10
General Labs (x2)	60 Workstations	PCs, Instructor’s Workstation	62
Cisco Network Lab	16 Workstations, 1 Instructor’s Workstation	PCs, Wi-Fi Access Point	18
Video Conference Room	1 Workstation, 1 Projector	PC, Projector, Wi-Fi	3
Lecturer’s Office	8 Workstations	PCs, Wi-Fi Access Point	9
Hybrid Classroom	1 Teacher Workstation, Video Conferencing Camera	Teacher PC, Camera, Wi-Fi	3
Embedded Lab	30 Workstations	PCs, IoT Devices, Wi-Fi	31
Server Room	Multiple Devices	Servers, Switches, Routers	10
		Total	153

Table 4.3.1: Number of Switch Ports Needed

Work Area	Number of Devices	Type of Devices	Number of Ports Required
Student Meeting Room	6 Workstations	PCs, Wi-Fi Access Point	7
Student Lounge	8 Workstations, 1 Printer	PCs, Printer, Wi-Fi	10
General Labs (x2)	60 Workstations	PCs, Instructor’s Workstation	62
Cisco Network Lab	16 Workstations, 1 Instructor’s Workstation	PCs, Wi-Fi Access Point	18
Video Conference Room	1 Workstation, 1 Projector	PC, Projector, Wi-Fi	3
Lecturer’s Office	8 Workstations	PCs, Wi-Fi Access Point	9

Hybrid Classroom	1 Teacher Workstation, Video Conferencing Camera	Teacher PC, Camera, Wi-Fi	3
Embedded Lab	30 Workstations	PCs, IoT Devices, Wi-Fi	31
Server Room	Multiple Devices	Servers, Switches, Routers	10
		Total	153

Table 4.3.2: Number of Patch Cords Needed

Work Area	Number of Devices	Type of Devices	Number of Ports Required
Student Meeting Room	6 Workstations	PCs, Wi-Fi Access Point	14
Student Lounge	8 Workstations, 1 Printer	PCs, Printer, Wi-Fi	20
General Labs (x2)	60 Workstations	PCs, Instructor's Workstation	124
Cisco Network Lab	16 Workstations, 1 Instructor's Workstation	PCs, Wi-Fi Access Point	36
Video Conference Room	1 Workstation, 1 Projector	PC, Projector, Wi-Fi	6
Lecturer's Office	8 Workstations	PCs, Wi-Fi Access Point	18
Hybrid Classroom	1 Teacher Workstation, Video Conferencing Camera	Teacher PC, Camera, Wi-Fi	6
Embedded Lab	30 Workstations	PCs, IoT Devices, Wi-Fi	62
Server Room	Multiple Devices	Servers, Switches, Routers	20
		Total	306

Table 4.3.3: Number of RJ45 Connector Needed

4.4 Identify cable types and length

Area	Length(m)	Total Price of Cable Usage
Ground Floor		
Room		
Student meeting room	10	200
Student lounge	23	460
General lab 1	54.5	1090
General lab 2	54.5	1090
Ground Floor Room Total		2840
First Floor		
Room		
Cisco Network Lab	62	1240
Video Conference Room	9	180
Lecturer office	17.5	350
Hybrid classroom	14.5	290
Embedded lab	53	1060
First Floor Room Total		3120

First Floor		
Room		
Path	Length(m)	Price of cables used(RM)
Server room to General Lab 1	80	1600
Server room to General Lab 2	90	1800
Server room to Student Lounge	83	1660
Ground Floor Room Total	261	5220
Total – First Floor		
Total length of wire to connect the room	261	5220
Total length of wire in different type	2840	56800

Total length of wire needed	3101	62020
Second Floor		
Server Room to Cisco Network Lab	38	760
Server Room to Video Conference Room	23	460
Server Room to Lecturer Office	17	340
Server Room to Hybrid classroom	39	780
Server Room to Embedded Lab	22	440
Second Floor Room Total	139	2780
Total - Second Floor		
Total length of wire to connect the room(m)	139	2780
Total length of wire in different type(m)	3120	62400
Total length of wire needed	3259	65180
Total (First Floor + Second Floor)	6360	127200

Explanation:

We detailed the distribution strategy and device connections. To enhance connectivity, selecting the appropriate cable types is essential. After thorough discussion, our team decided on a combination of CAT6 twisted pair cables and fiber optic cables for the building's network infrastructure.

CAT6 twisted pair cables will be used for both horizontal and vertical cabling in areas such as labs and lecture halls. It was because CAT6 can provide data transmission of as high as 10 Gbps due to the 550 MHz bandwidth. Moreover, the thicker copper wire allows for better dissipation of heat, which will enable the cable to perform under PoE, decreasing cabling complexity and cost. CAT6 cables are also the perfect fit for Ethernet applications within the building. We chose UTP CAT6 cables because they can reduce crosstalk and minimize electromagnetic interference from either natural or artificial sources.

Routers in the server room and connecting to an ISP will be connected using fiber optic cables. These cables use light to transmit data, enabling high transfer rates without degradation of the signal. The cables have a bandwidth of 10 Gbps, hence are suitable for handling massive volumes of data and maintaining quality signals over long distances.

In conclusion, the estimated total cable length of both first and second floor is 6360m. The estimated cable length will include both CAT6 and fiber optic cables that will ensure seamless internal connectivity between devices and a strong, reliable connection to the ISP.

Meeting Minutes

Meeting Minutes #1

Date/Time	14/12 December 2024 10:00pm			
Location Online	Google Meet			
Agenda	1. Overviewing task 4 2. Task distribution			
Meeting MC	Choh Jing Yi			
Attendance				
Name	Time	Reason For Absence		
Choh Jing Yi	22 00	-		
Neo Li Xin	22 00	-		
Pravinraj a/l Sivabathi	22 00	-		
Ahmad Ziyaad bin Mohd Abbas	22 00	-		
Meeting details				
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE	
1	Overviewing task 4 and make a simple discussion	-Choh Jing Yi share the task 4 information and rubric -All members read the task clearly and understand it	All members	
2	Suggest what to include in the diagram	-For the network distribution diagram and floor diagram, we discussed about the drawings criteria and how each network component should connect with each other	Choh Jing Yi and Neo Li Xin	
3	Task distribution for each group members	-Choh Jing Yi distribute task to each member equally. -Choh Jing Yi is tasked to create floor diagrams and room level network distribution diagram -Neo Li Xin is tasked to create network distribution	All members	

		diagram and overall floor diagram -Pravin is tasked to identify number of components in each room	
4	Meeting ended	We end our meeting at 10.30pm after all the discussion has been done.	All members

Meeting Minutes #2

Date/Time	16/12 December 2024 10:00pm
Location Online	Google Meet
Agenda	1. Review each member's current progress 2. Finalize network distribution diagram and floor plan 3. Verify the component information of each room
Meeting MC	Choh Jing Yi

Attendance

Name	Time	Reason For Absence
Choh Jing Yi	22 00	-
Neo Li Xin	22 00	-
Pravinraj a/l Sivabathi	22 00	-
Ahmad Ziyaad bin Mohd Abbas	22 00	-

Meeting details

NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Review of each member's progress	-Neo Li Xin and Choh Jing Yi shared their network distribution diagram and floor plan -Pravin shared the component information analysis of each room	All members
2	Finalize network distribution diagram	-Choh Jing Yi suggest creating another diagram for server room -Pravin suggested to use other elements to better represent the cable	Neo Li Xin
3	Finalize floor diagram	-Neo Li Xin suggested changing cable's element for better visibility	Choh Jing Yi

4	Finalize components details for each room	-Everyone made sure the cable length is sensible for each room to make sure the results are accurate	Pravin
5	Meeting ended	We end our meeting at 11.00pm after all the discussion has been done.	All members